Universitat Pompeu Fabra Barcelona

MTG Music Technology Group

Music Technology Lab S103

An Interface for Generative Audio Latent Interpolation

Esteban Gutiérrez

Course Structure

Class 1. Course Introduction Part 1 ✅
Class 2. Course Introduction Part 2 ✅
Class 3. Project Introduction Part 1: Background + Dataset Curation ✅
Class 4. Project Introduction Part 2: Training SCAPES + Project Details ⬅️
Class 5 and so on. Project Development…

Generative Models

A generative model is a type of machine learning model that learns the underlying patterns or probability distribution of data to generate new, realistic examples.

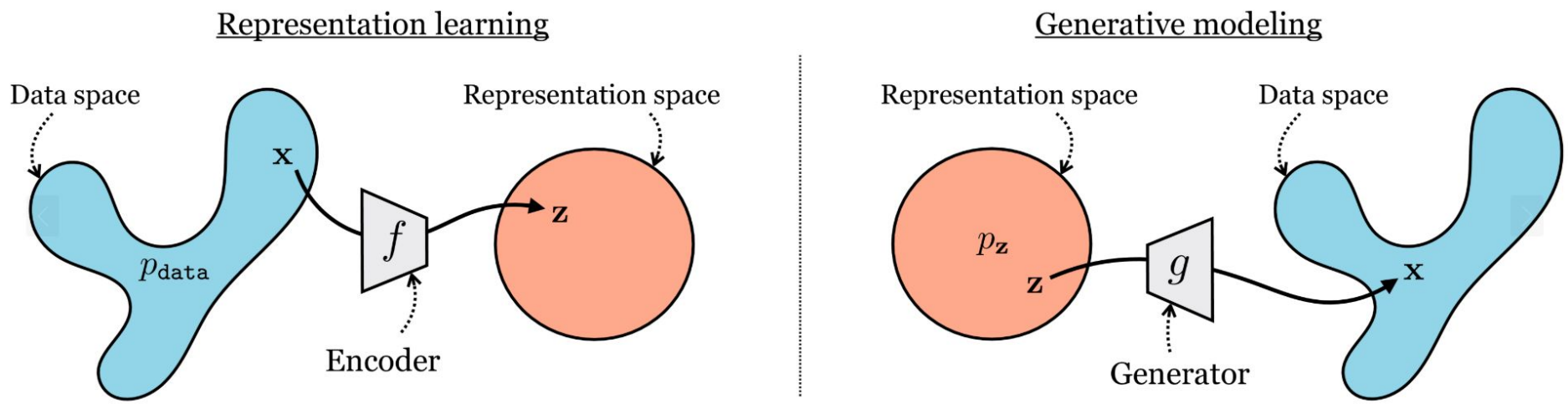
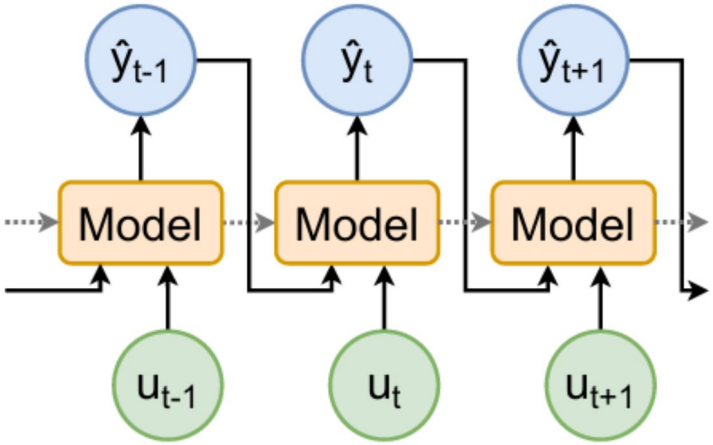
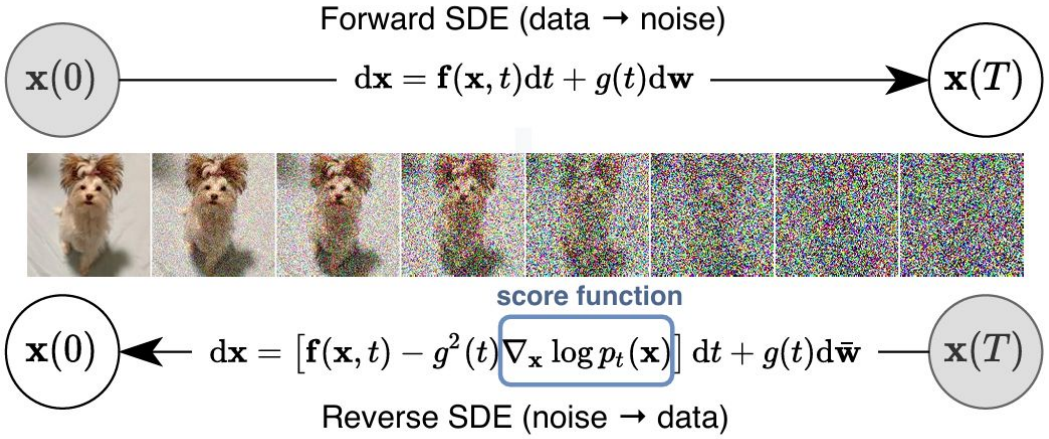
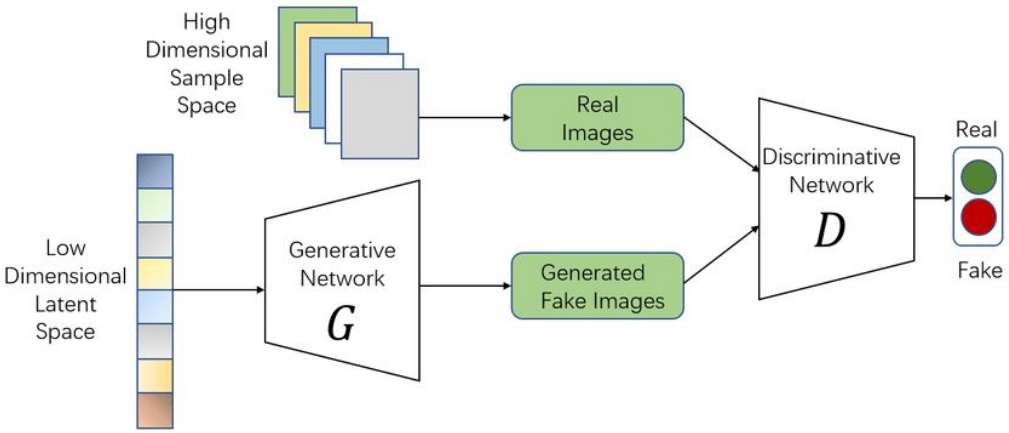


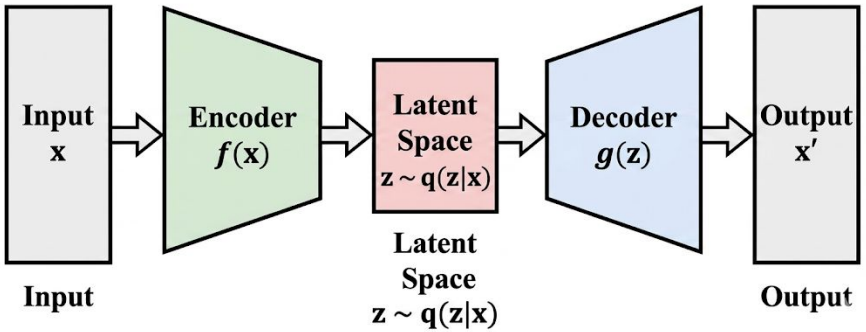
Figure from *"Foundation of Computer Vision" online book*

Generative Models









Figures from “Non-Autoregressive vs Autoregressive Neural Networks for System Identification” (Weber & Guhmann) and “What Are Generative Adversarial Networks (GANs)? Understanding and Implications” (Babba, S.)

Neural Ordinary Differential Equations

Neural Ordinary Differential Equations are a change in the paradigm, where the *flow* of the information is modeled as a single vector field.

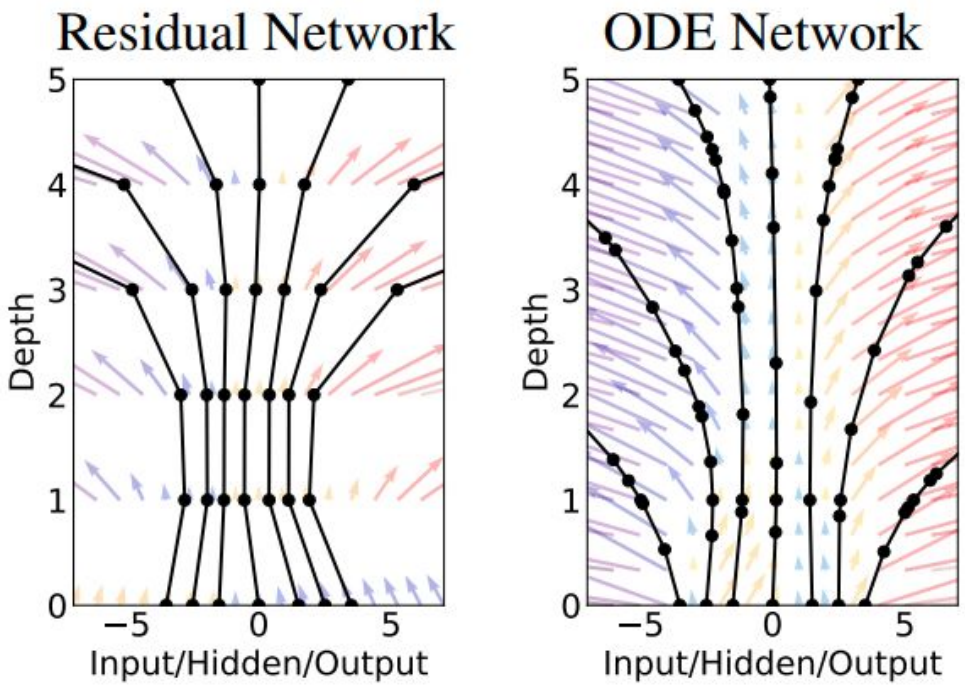
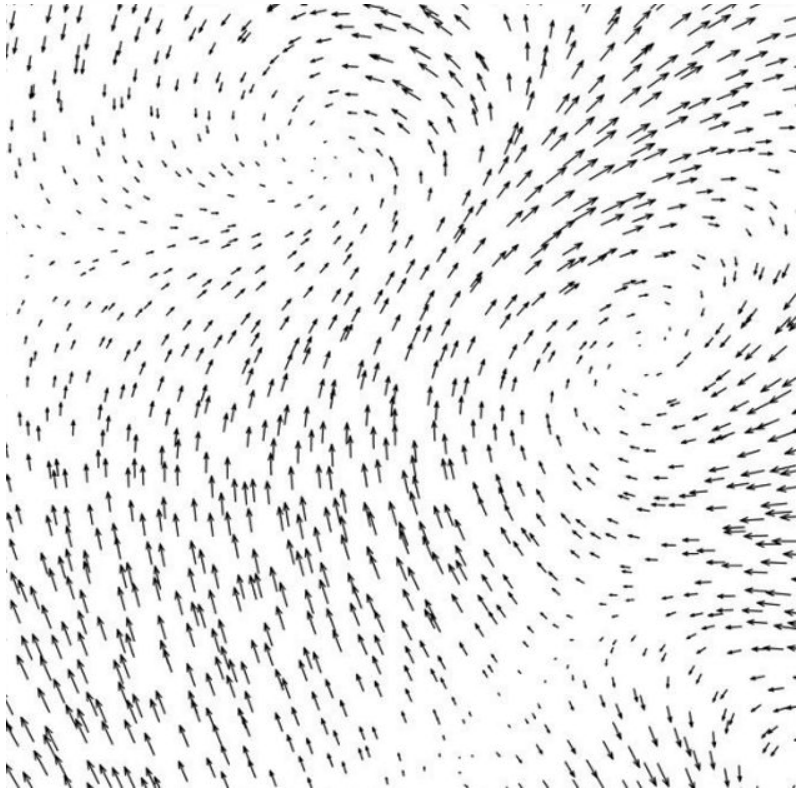


Figure 1: *Left:* A Residual network defines a discrete sequence of finite transformations. *Right:* A ODE network defines a vector field, which continuously transforms the state. *Both:* Circles represent evaluation locations.

Neural Ordinary Differential Equations

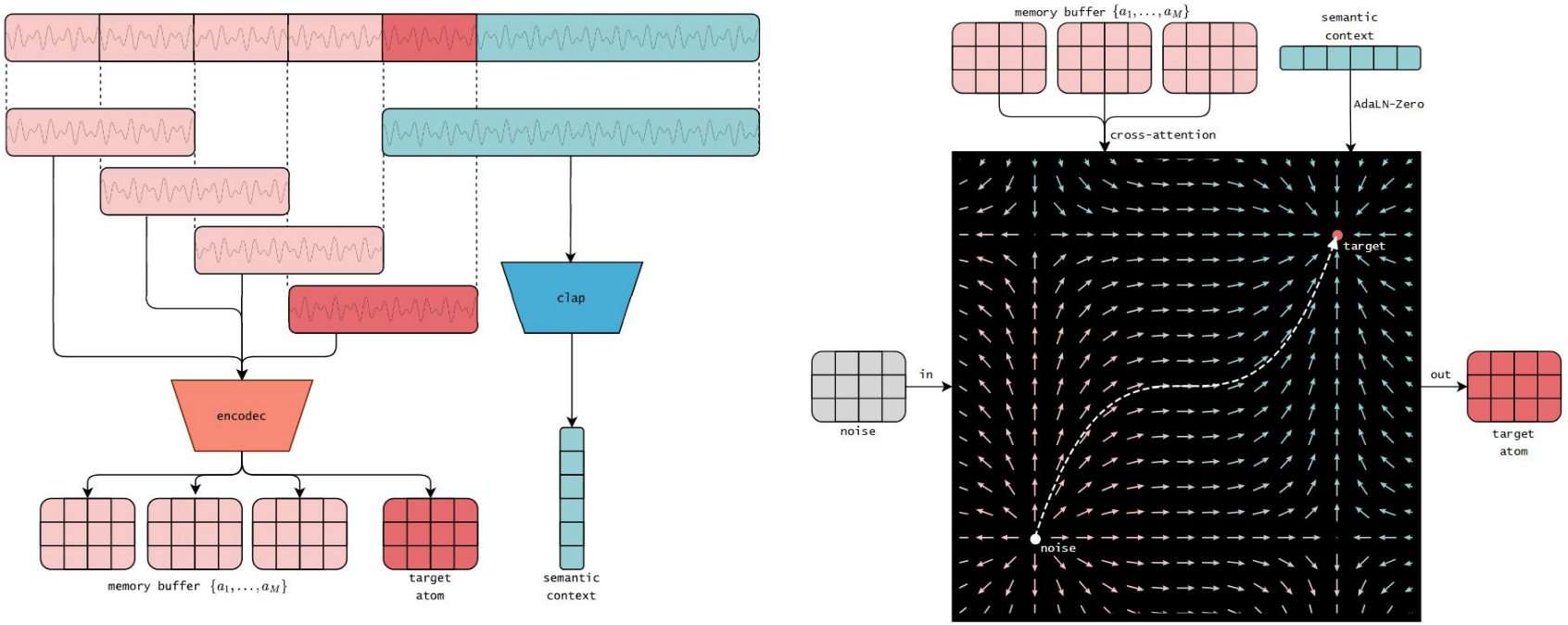
A Continuous Normalizing Flow is a model based on an ODE network where the vector field also depends in the time parameter. In such a case, the generation map correspond to solving a particular ODE.

$$\begin{cases} \dfrac{dx(t)}{dt} & = & F(x(t), \theta(t)) \\ x(0) & = & x_0 \end{cases}$$



SCAPES

SCAPES: Semantically Conditioned Autoregressive Prior for Environmental Sounds, is a generative model trained without manual annotations, designed for high-fidelity, semantically controllable stereo sound synthesis with minimal data and computational cost. SCAPES architecture is a small-scale (CNF) that serves as a conditional prior over EnCodec latent representations. SCAPES applies its architecture sequentially over time, conditioning each segment on a memory buffer of previous latent states and high-level semantic embeddings from CLAP.



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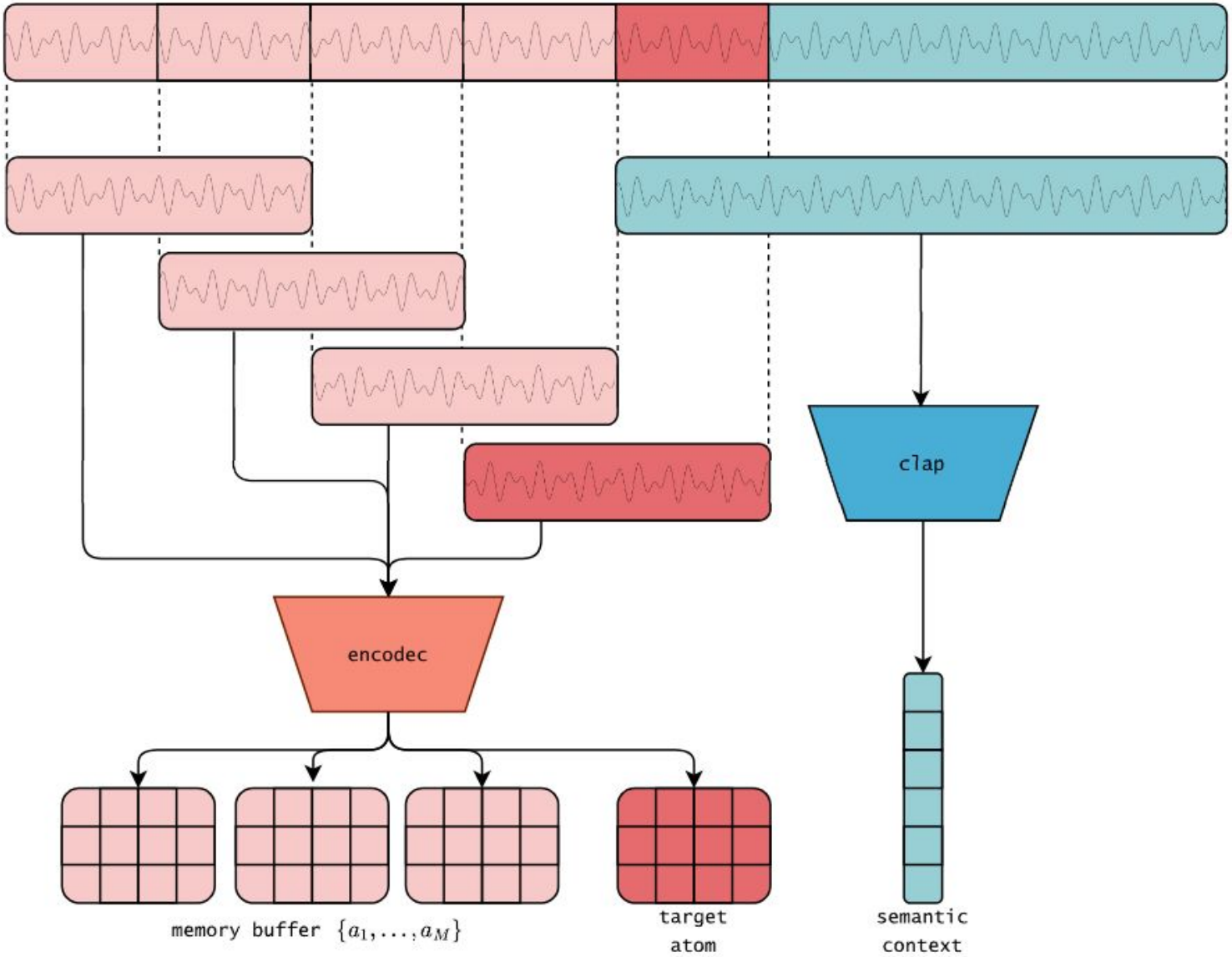
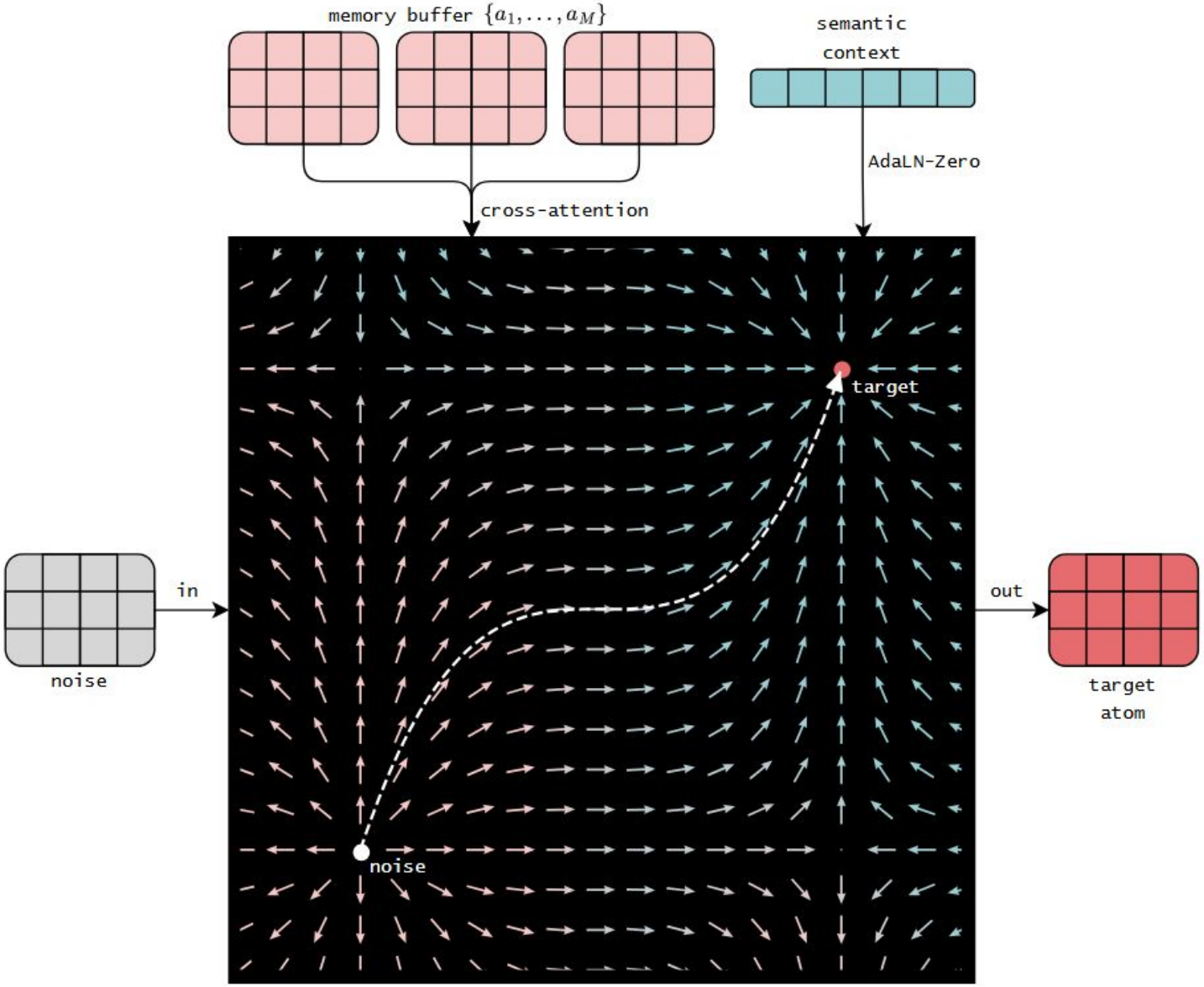
Properties:

1. Trains extremely fast in comparison to other generative models.
2. The main architecture is very small in comparison to other generative models.
3. It predicts EnCodec latents, so it is stereo by default
4. It doesn't need annotations since it uses automatically computed CLAP embeddings.
5. It is based on a CNF, so it is stochastic.
6. Designed for sounds that do not have a strong time structure.

SCAPES Data Preparation

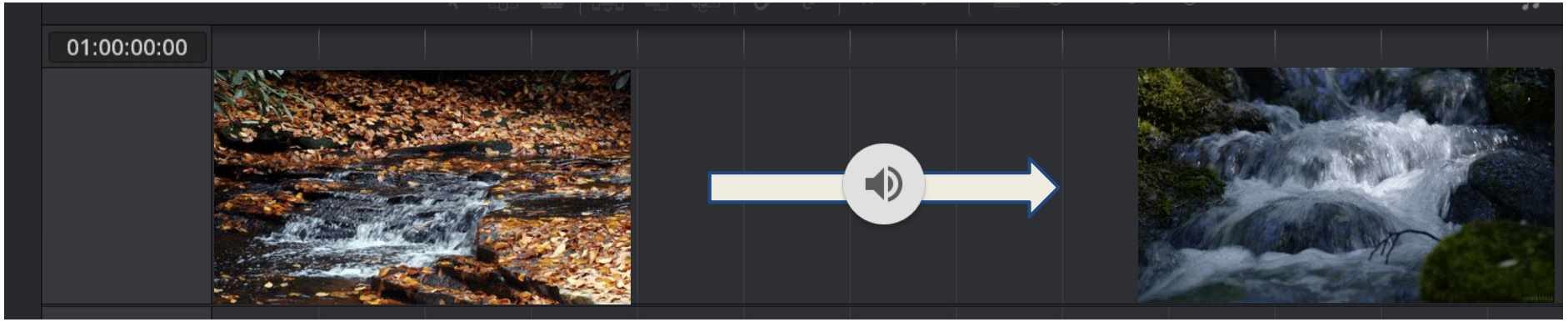


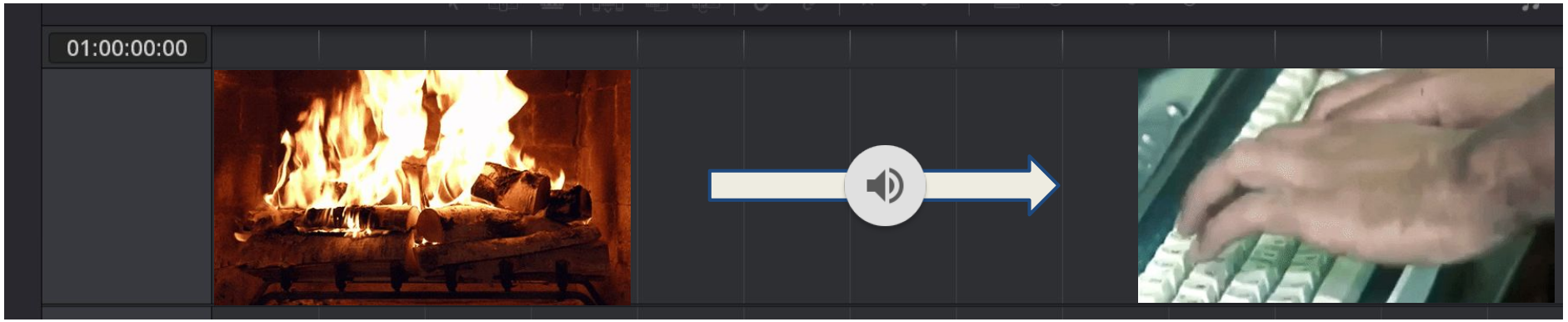
SCAPES Inference



SCAPES Synthesis Examples

SCAPES Interpolation Examples





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SCAPES backend:

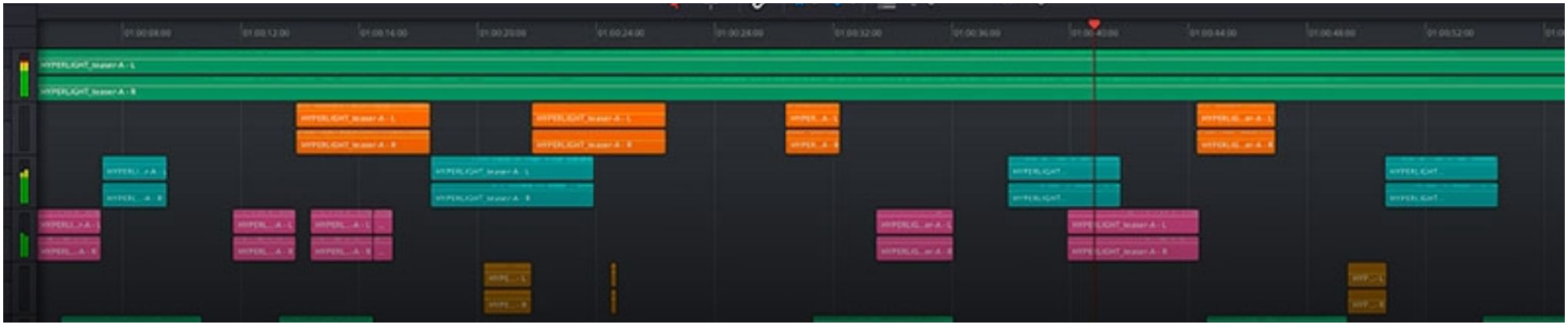
1. Curate a dataset ✅
2. Train SCAPES and test it. If it doesn't work properly, go back to step 1. ⬅️

Interpolation code:

1. Study the interpolation code. ⬅️
2. Implement the main interpolation function.

Interpolation GUI:

1. Study GUI possibilities. ⬅️
2. Study deployment possibilities. ⬅️
3. Implement GUI.



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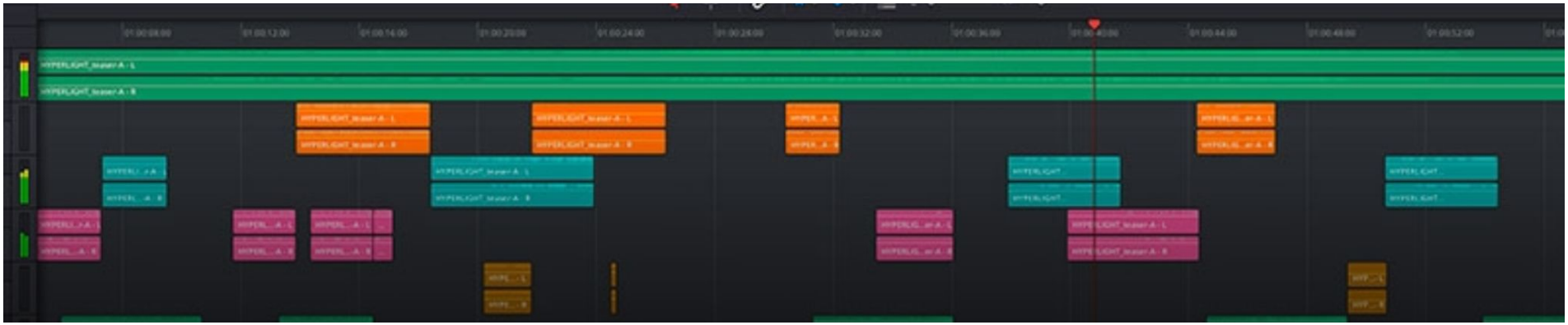
Project Requirements:

1. It must be inside a GitHub repository.
2. It must clone SCAPES to be used, not just copy it to the repo.
3. It must run a GUI where the user can build its own timeline.
4. The user must be able to drag and drop parts of the training data to the timeline.
5. It must be deployed to the public by the end of the term.

Recommendations:

1. Study well the deployment options before start coding the GUI.
2. I suggest deploying on Hugging Face or Streamlit Community Cloud.
3. Start with one track and then extend to multiple tracks.
4. Study well how to do inference using SCAPES.



Thanks for listening!